

# Alireza Fadaei

Python Backend · Applied AI · Secure Software Systems

Software engineer and researcher based in Oulu, Finland. Open to remote opportunities.

alizfadaei98@gmail.com

+358 41 580 0895

arfadaei.ir · GitHub

LinkedIn · Oulu, Finland

## EXPERIENCE

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| 2026      | <b>Software Engineer · Aminyx (Remote)</b><br>Built agentic AI features for automated sourcing, multi-layered trust-scoring systems, and secure infrastructure for B2B trade workflows.                   |
| 2024–2026 | <b>Programming Instructor · Iran TVTO</b><br>Taught programming fundamentals and software engineering practices through hands-on, industry-aligned projects.                                              |
| 2024–2026 | <b>Chair, Computer Department · Kabir Technical and Vocational School</b><br>Led the department team and developed curricula in programming, network security, and machine learning.                      |
| 2022–2024 | <b>Software Developer · Fardaad (ISTT)</b><br>Built vulnerability and patch-management tools for municipal and government networks and improved backend algorithms supporting patch deployment workflows. |
| 2021–2022 | <b>Backend Developer &amp; Server Administrator · BrassicaLNC / PCG Lab</b><br>Built and administered a biomedical information platform covering data extracted from 2,361 public RNA-Seq libraries.      |
| 2021–2022 | <b>Backend Developer &amp; Server Administrator · MetaMenu</b><br>Contributed to a cloud-based restaurant platform supporting interactive menus, ordering, delivery, and accounting workflows.            |

## EDUCATION

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| 2026–Present | <b>M.Sc. Software Engineering and Information Systems</b><br>University of Oulu, Finland |
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| 2022–2025 | <b>M.Sc. Software Engineering</b><br>Islamic Azad University, Najafabad Branch · Top student · Thesis grade: 20/20<br>Thesis: Optimized Federated Deep Learning for Intrusion Detection in Dew Computing Environments |
| 2017–2022 | <b>B.Sc. Computer Engineering</b><br>University of Isfahan · Best B.Sc. Thesis Award<br>Thesis: Design and Implementation of an ADS-B-Based Aircraft Observation System                                               |

## SELECTED PUBLICATION

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| 2026 | <b>A Novel Intrusion Detection System for Dew Computing Environments Based on an Enhanced Federated Deep Learning Model</b><br>A. Fadaei and B. Barekatin · <i>The Journal of Supercomputing</i> , Springer · DOI |
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## TECHNICAL SKILLS

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| Backend               | Python, Django, Flask, FastAPI, REST APIs, Celery, Node.js               |
| Frontend (learning)   | React, modern JavaScript, component-based and responsive interfaces      |
| AI & Data             | TensorFlow, Scikit-learn, Pandas, NumPy, Federated Learning              |
| Data & Infrastructure | PostgreSQL, Redis, Neo4j, Docker, Nginx, Linux, VMware                   |
| Security              | Vulnerability Management, Intrusion Detection, AI Security, IoT Security |
| Collaboration         | Git, code review, technical documentation, asynchronous communication    |

## ADDITIONAL

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**English:** IELTS 7.5 (Reading 8.5, Listening 7.5, Writing 7.0, Speaking 6.5)  
**Certification:** LPIC-1 · **Timezone:** EET/EEST